

Bundle 12 — Paperclip-Agentfield adapter

Scrum-master view

Bundle objective: create the bridge between **Paperclip** as the human-facing workflow/dashboard and **Agentfield** as the experiment-aware controller.

Bundle 11 established the first Agentfield pattern:

```
GRNExperiment intent
-> ExperimentAwareController
-> stage resolver
-> agent registry
-> reasoner invoker
-> selected GRN reasoners
-> status/results
-> Agentfield UI DAG
```

Bundle 12 now prepares the adapter that lets Paperclip work with that lifecycle.

The adapter should answer:

Can Paperclip request a GRN experiment without knowing Agentfield internals?

Can Paperclip display Agentfield experiment status, selected agents, stage results, artifact refs, failures, and review-needed states?

Can a human approve, reject, retry, annotate, or escalate experiment outputs from Paperclip?

Can Paperclip eventually launch NCA-ART-GRN mechanism-discovery work through Agentfield, without becoming the controller itself?

The core rule:

Paperclip is the human workflow surface.
Agentfield is the experiment controller.
nca-art-grn is the science engine.
config prepares the environment.

Where this bundle sits in the platform

config
prepares adapter workspace, schemas, fixtures, mappings, smoke checks

Paperclip-Agentfield adapter
maps Paperclip jobs/actions to Agentfield execute requests
maps Agentfield status/results back to Paperclip-visible objects

Agentfield
owns run_experiment and experiment lifecycle

nca-art-grn
later owns real DSL/PDE/NCA/ART2/ARTMAP research execution

Paperclip
owns dashboard, inbox, review, approvals, user workflow, visibility

Bundle 12 is **not** the Paperclip app itself.
It is **not** Agentfield itself.
It is the translation and integration layer between them.

User story

As the Operator / Research Scientist,
I want Paperclip to submit and review Agentfield GRN experiments through a clear adapter,
so that I can start experiment-aware workflows from a human-facing dashboard, inspect
status/results/artifacts, and approve or reject next actions without manually calling
Agentfield endpoints.

Product outcome

After Bundle 12, this Paperclip-side request:

Create a GRN discovery experiment for this dataset using PCA, correlation,
perturbation planning, and hypothesis ranking.

should map to the Agentfield execute shape from the Bundle 11 POC:

```
POST /api/v1/execute/grn-experiment.run_experiment
```

```
{
  "input": {
    "name": "GRN Discovery in Human Cortex",
    "description": "Identify key transcription factor regulatory networks in human cortical development u",
    "dataset_ref": "GSE123456_cortex_scrna",
    "organism": "human",
    "method_flags": ["pca", "correlation", "perturbation"]
  }
}
```

Then the adapter maps Agentfield's response/status into a Paperclip-visible object:

PaperclipExperimentCard:

- title
- phase
- selected agents
- stage summaries
- execution id
- artifact links
- failure reason
- review actions

Step type legend

CONFIG STEP

Managed bootstrap/check/smoke step invoked by config.

ADAPTER CODE

Code inside /workspace/repos/paperclip-agentfield-adapter.

PAPERCLIP CONTRACT

Job/action/review schema visible to Paperclip.

AGENTFIELD CONTRACT

Execute request, execution id, status/result shape from Agentfield.

RESEARCH OUTPUT

Mechanism reports, search reports, artifacts, stage summaries.

AGENTFIELD LATER

Already started in Bundle 11; Bundle 12 calls it.

Concretizations / managed steps

```
prepare_paperclip_agentfield_adapter_workspace  
prepare_paperclip_job_schema  
prepare_agentfield_execute_client  
prepare_agentfield_request_mapper  
prepare_agentfield_status_mapper  
prepare_artifact_link_mapper  
prepare_review_action_mapper  
prepare_adapter_config_profiles  
prepare_adapter_smoke_fixtures  
prepare_adapter_cli_smoke_commands  
check_paperclip_agentfield_adapter_ready  
run_paperclip_agentfield_adapter_dryrun_smoke
```

Optional live step, only when Agentfield server is running:

```
run_paperclip_agentfield_adapter_live_smoke
```

Not first pass:

```
submit_real_paperclip_job  
write_to_paperclip_database  
launch_runpod_from_paperclip  
auto_approve_next_experiment
```

What each step should do

prepare_paperclip_agentfield_adapter_workspace

Type: CONFIG STEP

Owner: config prepares workspace.

Target role: usually aiengineer .

Creates:

```
/workspace/repos/paperclip-agentfield-adapter/  
  README.md  
  pyproject.toml  
  paperclip_agentfield_adapter/  
    __init__.py  
    schemas/  
    clients/  
    mappers/  
    review/  
    artifacts/  
    cli/  
  configs/  
    adapter.yaml  
    agentfield_endpoints.yaml  
    paperclip_profiles.yaml  
  fixtures/  
    paperclip_jobs/  
    agentfield_responses/  
  smoke_tests/  
  runs/
```

What this does for the platform:

Creates a clean adapter repo, separate from Agentfield and separate from Paperclip.

What this does for the research:

Gives the experiment workflow a human-review bridge without polluting the scientific repo or Agentfield controller code.

How it links into the whole system:

```
config creates adapter workspace
adapter calls Agentfield
Paperclip later calls adapter
Agentfield calls reasoners/research engine
Paperclip receives reviewable results
```

prepare_paperclip_job_schema

Type: CONFIG STEP

Creates/validates ADAPTER CODE + PAPERCLIP CONTRACT:

```
paperclip_agentfield_adapter/schemas/paperclip_job.py
configs/paperclip_job.schema.yaml
```

Paperclip job schema must encode:

```
job_id
title
description
job_type
requested_by
created_at
dataset_ref
organism
intent
method_flags
research_mode
candidate_id
candidate_batch_id
review_policy
priority
```

Initial supported `job_type` values:

```
grn_experiment_poc  
nca_art_mechanism_smoke  
parameter_search_review  
mechanism_report_review
```

What this does for the platform:

Defines what Paperclip can ask for without knowing Agentfield internals.

What this does for the research:

Allows the Research Scientist to request experiments in research language: dataset, mechanism intent, candidate, method flags, review policy.

How it links into the whole system:

```
Paperclip job -> adapter schema validation  
adapter -> Agentfield request mapper  
Agentfield -> run_experiment
```

prepare_agentfield_execute_client

Type: CONFIG STEP

Creates ADAPTER CODE:

```
paperclip_agentfield_adapter/clients/agentfield_client.py  
configs/agentfield_endpoints.yaml
```

Client must support:

```
sync execute request  
async execute request  
poll execution status
```


fetch execution result
timeout handling
server health check

Initial endpoint profile:

```
agentfield:  
  base_url: "http://localhost:8080"  
  execute_sync_path: "/api/v1/execute/{node}.{reasoner}"  
  execute_async_path: "/api/v1/execute_async/{node}.{reasoner}"  
  default_node: "grn-experiment"  
  default_reasoner: "run_experiment"
```

The developer POC trigger uses the sync execute endpoint for `grn-experiment.run_experiment`, while noting that Agentfield also has an async execute endpoint for longer workflows.

What this does for the platform:

Gives the adapter one clean way to call Agentfield.

What this does for the research:

Lets research workflows become dashboard-triggerable without manually writing `curl` commands.

How it links into the whole system:

```
Paperclip job -> Agentfield client -> Agentfield execute endpoint  
Agentfield response -> status mapper -> Paperclip card
```

`prepare_agentfield_request_mapper`

Type: CONFIG STEP

Creates ADAPTER CODE:

```
paperclip_agentfield_adapter/mappers/request_mapper.py  
configs/request_mapping.yaml
```

The mapper converts Paperclip job fields into Agentfield input.

For the current POC:

```
PaperclipJob.title      -> input.name  
PaperclipJob.description -> input.description  
PaperclipJob.dataset_ref -> input.dataset_ref  
PaperclipJob.organism   -> input.organism  
PaperclipJob.method_flags -> input.method_flags
```

Example mapping:

```
{  
  "input": {  
    "name": "{title}",  
    "description": "{description}",  
    "dataset_ref": "{dataset_ref}",  
    "organism": "{organism}",  
    "method_flags": "{method_flags}"  
  }  
}
```

Future fields should pass through when Bundle 11 schema supports them:

```
research_mode  
candidate_id  
candidate_batch_id  
mechanism_hypothesis_id
```

`config_ref`**What this does for the platform:**

Decouples Paperclip's job model from Agentfield's reasoner input model.

What this does for the research:

Preserves experiment meaning during translation. The adapter should not collapse a mechanism-discovery request into a generic prompt.

How it links into the whole system:

```
Paperclip job -> request mapper -> Agentfield GRNExperiment input
```

`prepare_agentfield_status_mapper`

Type: CONFIG STEP

Creates ADAPTER CODE:

```
paperclip_agentfield_adapter/mappers/status_mapper.py  
configs/status_mapping.yaml
```

The mapper converts Agentfield status/results into Paperclip display fields.

Current POC output shape includes:

```
phase  
selected_agents  
stage_results  
final_summary
```

The POC `GRNExperimentStatus` accumulates selected agents, per-stage results, and a final summary.

Paperclip-visible status should include:

```
job_id  
execution_id  
phase  
selected_agents  
stage_count  
completed_stage_count  
stage_summaries  
final_summary  
failure_reason  
review_required
```

Phase mapping:

```
Pending -> queued  
Running -> running  
Completed -> completed_review_required  
Failed -> failed
```

What this does for the platform:

Turns Agentfield internal status into human-facing workflow status.

What this does for the research:

Lets the user see which reasoning/science stages ran and what they concluded.

How it links into the whole system:

```
Agentfield result -> status mapper -> Paperclip experiment card
```

prepare_artifact_link_mapper

Type: CONFIG STEP**Creates ADAPTER CODE:**

```
paperclip_agentfield_adapter/mappers/artifact_mapper.py  
configs/artifact_mapping.yaml
```

First-pass POC artifacts:

```
stage summaries  
final summary  
execution id  
DAG/UI link
```

Future NCA-ART-GRN artifacts:

```
mechanism_report.md  
search_report.md  
candidate.dsl.json  
pattern_dynamics.json  
nca_summary.json  
art2_prototypes.json  
artmap_transitions.json  
perturbation_summary.json  
figures  
tables  
failure_reason.txt
```

Each artifact link should encode:

```
artifact_type  
label  
path_or_url  
required  
exists
```

```
display_priority  
review_action_hint
```

What this does for the platform:

Makes outputs visible and reviewable instead of buried in run folders.

What this does for the research:

Surfaces the actual evidence chain: mechanism reports, NCA/ART/ARTMAP outputs, perturbation summaries, and candidate DSL files.

How it links into the whole system:

Agentfield status/artifact refs -> artifact mapper -> Paperclip visible links

prepare_review_action_mapper

Type: CONFIG STEP

Creates ADAPTER CODE:

```
paperclip_agentfield_adapter/review/actions.py  
configs/review_actions.yaml
```

Review actions:

```
approve_result  
reject_result  
request_retry  
request_more_evidence  
promote_to_mechanism_candidate  
promote_to_zettelkasten_alloy  
queue_next_experiment  
mark_as_failed_science  
mark_as_platform_failure
```

First pass should not execute all actions. It should define the schema and dry-run mappings.

Each action must encode:

```
action_id
label
requires_human
allowed_phases
payload_schema
side_effects
dryrun_only_first_pass
```

What this does for the platform:

Defines how humans interact with Agentfield results through Paperclip.

What this does for the research:

Keeps the Research Scientist in control. The system can suggest next experiments, but a human approves before costly or publication-relevant work.

How it links into the whole system:

```
Paperclip review action -> adapter action mapper
later -> Agentfield new experiment or PKM bridge
```

prepare_adapter_config_profiles

Type: CONFIG STEP

Creates:

```
configs/adapter.yaml
configs/paperclip_profiles.yaml
configs/agentfield_endpoints.yaml
```

Profiles:

```
local_dev  
agentfield_poc  
nca_art_grn_dev  
paperclip_mock
```

Must encode:

```
agentfield_base_url  
default_node  
default_reasoner  
sync_or_async  
poll_interval  
timeout_seconds  
paperclip_mock_mode  
artifact_base_path  
review_policy
```

What this does for the platform:

Lets the same adapter run in local dev, mock mode, or later real Paperclip mode.

What this does for the research:

Allows safe iteration without accidentally submitting real workflows or calling remote systems.

How it links into the whole system:

```
config profile -> adapter behavior  
local smoke -> mock mode  
live smoke -> Agentfield local server  
future -> Paperclip integration
```


prepare_adapter_smoke_fixtures

Type: CONFIG STEP

Creates fixtures:

```
fixtures/paperclip_jobs/grn_experiment_poc.json  
fixtures/paperclip_jobs/nca_art_mechanism_smoke.json  
fixtures/agentfield_responses/grn_experiment_completed.json  
fixtures/agentfield_responses/grn_experiment_failed.json
```

Current POC Paperclip job fixture should map to the developer's example:

```
{  
  "job_id": "pc-grn-poc-001",  
  "title": "GRN Discovery in Human Cortex",  
  "description": "Identify key transcription factor regulatory networks in human cortical development using scRNA-seq",  
  "job_type": "grn_experiment_poc",  
  "dataset_ref": "GSE123456_cortex_scrna",  
  "organism": "human",  
  "method_flags": ["pca", "correlation", "perturbation"],  
  "review_policy": "human_required"  
}
```

What this does for the platform:

Gives deterministic test data.

What this does for the research:

Lets you validate the adapter path before real experiments or real Paperclip integration.

How it links into the whole system:

```
fixture Paperclip job -> request mapper -> Agentfield request  
fixture Agentfield response -> status mapper -> Paperclip card
```

prepare_adapter_cli_smoke_commands

Type: CONFIG STEP

Creates CLI/script wrappers:

```
paperclip_agentfield_adapter/cli/dryrun.py  
paperclip_agentfield_adapter/cli/live_smoke.py  
smoke_tests/run_adapter_dryrun.sh  
smoke_tests/run_adapter_live_smoke.sh
```

Dryrun command should:

```
load Paperclip job fixture  
validate Paperclip schema  
map to Agentfield execute request  
load mock Agentfield response  
map to Paperclip status/card  
write dryrun output
```

Live smoke command should:

```
load fixture  
map to Agentfield request  
call local Agentfield execute endpoint  
map response  
write live output
```

What this does for the platform:

Makes adapter behavior testable without the full Paperclip app.

What this does for the research:

Lets you verify the whole "human request -> Agentfield experiment -> human-readable result" path.

How it links into the whole system:

CLI dryrun -> no server required
CLI live -> Agentfield server required
future Paperclip -> same adapter library

`check_paperclip_agentfield_adapter_ready`

Type: CONFIG CHECK STEP

Runs: no Agentfield call, no Paperclip call.

Should check:

adapter workspace exists
schemas exist
client exists
request mapper exists
status mapper exists
artifact mapper exists
review action mapper exists
fixtures exist
configs exist
dryrun CLI exists
Python imports available
Agentfield URL configured or defaultable

What this does for the platform:

Preflight before dryrun/live testing.

What this does for the research:

Prevents adapter wiring issues from being confused with Agentfield or science failures.

How it links into the whole system:

```
aiengineer runs check  
fixes adapter setup  
then runs dryrun or live smoke
```

run_paperclip_agentfield_adapter_dryrun_smoke

Type: CONFIG SMOKE STEP

Runs: no Agentfield server required.

Should do:

```
load Paperclip job fixture  
validate job schema  
map to Agentfield execute request  
load mock Agentfield completed response  
map to Paperclip card/status  
map mock artifacts  
map review actions  
write dryrun report
```

Output:

```
/workspace/runs/paperclip-agentfield-adapter/dryrun/<timestamp>/  
paperclip_job.json  
agentfield_request.json  
mock_agentfield_response.json  
paperclip_status.json
```

```
paperclip_card.json  
review_actions.json  
dryrun_report.md
```

What this does for the platform:

Proves the adapter translation works without needing live systems.

What this does for the research:

Confirms that experiment intent and stage evidence survive translation.

How it links into the whole system:

```
today:  
  mock Paperclip job -> adapter dryrun  
  
later:  
  Paperclip UI -> adapter -> Agentfield
```

run_paperclip_agentfield_adapter_live_smoke

Type: CONFIG SMOKE STEP

Runs: live local Agentfield call only when explicitly configured.

Requires:

```
af server running  
grn_experiment.py Agentfield node running  
OPENROUTER_API_KEY set if using live POC model calls  
AGENTFIELD_URL reachable
```

Should do:

```
load Paperclip fixture
map to Agentfield execute request
call /api/v1/execute/grn-experiment.run_experiment
capture response
map to Paperclip status/card
write live smoke output
```

Output:

```
/workspace/runs/paperclip-agentfield-adapter/live/<timestamp>/
paperclip_job.json
agentfield_request.json
agentfield_response.json
paperclip_status.json
paperclip_card.json
```

What this does for the platform:

Proves the adapter can call the current Bundle 11 POC.

What this does for the research:

Shows that a dashboard-style request can trigger the experiment-aware controller.

How it links into the whole system:

```
Paperclip-like job -> adapter -> Agentfield POC -> status/card
```

Whole-system linkage

Bundle 12 connects like this:

Bundle 11

Agentfield POC/controller exposes run_experiment

Bundle 12

adapter maps Paperclip job to run_experiment input

adapter maps Agentfield result to Paperclip status/card

Bundle 13

later uses this bridge for full GRN discovery campaigns

Paperclip

later becomes the actual UI/inbox/review system

Concrete flow:**Paperclip job:**

"Run GRN experiment"

Adapter:

validate job

map request

call Agentfield

map status/results

expose review actions

Agentfield:

run_experiment

selected agents

stage results

final summary

Paperclip:

show card

ask human for review action

Acceptance criteria

The following should become valid:

```
sudo config --target aiengineer bootstrap step prepare_paperclip_agentfield_adapter_workspace
sudo config --target aiengineer bootstrap step prepare_paperclip_job_schema
sudo config --target aiengineer bootstrap step prepare_agentfield_execute_client
sudo config --target aiengineer bootstrap step prepare_agentfield_request_mapper
sudo config --target aiengineer bootstrap step prepare_agentfield_status_mapper
sudo config --target aiengineer bootstrap step prepare_artifact_link_mapper
sudo config --target aiengineer bootstrap step prepare_review_action_mapper
sudo config --target aiengineer bootstrap step prepare_adapter_config_profiles
sudo config --target aiengineer bootstrap step prepare_adapter_smoke_fixtures
sudo config --target aiengineer bootstrap step prepare_adapter_cli_smoke_commands
sudo config --target aiengineer bootstrap step check_paperclip_agentfield_adapter_ready
```

Explicit dryrun:

```
sudo config --target aiengineer bootstrap step run_paperclip_agentfield_adapter_dryrun_smoke
```

Optional explicit live smoke:

```
sudo config --target aiengineer bootstrap step run_paperclip_agentfield_adapter_live_smoke
```

Prepare/check steps must:

```
not require real Paperclip
not require live Agentfield server
not call OpenRouter
not print API keys
not launch experiments
```



```
not launch Runpod
not write to Paperclip database
not auto-approve next actions
```

Dryrun smoke must:

```
use fixtures only
write mapped request/status/card
not call Agentfield
not call Paperclip
not call model APIs
```

Live smoke must:

```
require explicit live mode
call only local Agentfield endpoint
record request/response
not submit real Paperclip data
not call Runpod
```

Proposed tests

Registry and syntax tests

```
bash --noprofile --norc -n /home/vmuser/.local/bin/config.sh
bash --noprofile --norc -n /home/vmuser/.local/lib/config-sh/installers.sh

config bootstrap steps | grep prepare_paperclip_agentfield_adapter_workspace
config bootstrap steps | grep prepare_paperclip_job_schema
config bootstrap steps | grep prepare_agentfield_execute_client
config bootstrap steps | grep prepare_agentfield_request_mapper
```

```
config bootstrap steps | grep prepare_agentfield_status_mapper  
config bootstrap steps | grep run paperclip_agentfield_adapter_dryrun_smoke
```

Workspace tests

```
sudo config --target aiengineer bootstrap step prepare_paperclip_agentfield_adapter_workspace  
  
test -d /workspace/repos/paperclip-agentfield-adapter/paperclip_agentfield_adapter/schemas  
test -d /workspace/repos/paperclip-agentfield-adapter/paperclip_agentfield_adapter/clients  
test -d /workspace/repos/paperclip-agentfield-adapter/paperclip_agentfield_adapter/mappers  
test -d /workspace/repos/paperclip-agentfield-adapter/paperclip_agentfield_adapter/review  
test -d /workspace/repos/paperclip-agentfield-adapter/fixtures/paperclip_jobs  
test -d /workspace/repos/paperclip-agentfield-adapter/fixtures/agentfield_responses
```

Schema tests

```
sudo config --target aiengineer bootstrap step prepare_paperclip_job_schema  
  
test -f /workspace/repos/paperclip-agentfield-adapter/configs/paperclip_job.schema.yaml  
test -f /workspace/repos/paperclip-agentfield-adapter/paperclip_agentfield_adapter/schemas/paperclip_job
```

Schema must include:

```
job_id  
title  
description  
job_type  
dataset_ref  
organism  
method_flags  
review_policy
```

Client tests

```
sudo config --target aiengineer bootstrap step prepare_agentfield_execute_client

test -f /workspace/repos/paperclip-agentfield-adapter/paperclip_agentfield_adapter/clients/agentfield_cli:
test -f /workspace/repos/paperclip-agentfield-adapter/configs/agentfield_endpoints.yaml
grep "grn-experiment" /workspace/repos/paperclip-agentfield-adapter/configs/agentfield_endpoints.yaml
grep "run_experiment" /workspace/repos/paperclip-agentfield-adapter/configs/agentfield_endpoints.yaml
```

Mapper tests

```
sudo config --target aiengineer bootstrap step prepare_agentfield_request_mapper
sudo config --target aiengineer bootstrap step prepare_agentfield_status_mapper
sudo config --target aiengineer bootstrap step prepare_artifact_link_mapper

test -f /workspace/repos/paperclip-agentfield-adapter/paperclip_agentfield_adapter/mappers/request_mapper:
test -f /workspace/repos/paperclip-agentfield-adapter/paperclip_agentfield_adapter/mappers/status_mapper:
test -f /workspace/repos/paperclip-agentfield-adapter/paperclip_agentfield_adapter/mappers/artifact_mapper:
```

Fixture tests

```
sudo config --target aiengineer bootstrap step prepare_adapter_smoke_fixtures

test -f /workspace/repos/paperclip-agentfield-adapter/fixtures/paperclip_jobs/grn_experiment_poc.json
test -f /workspace/repos/paperclip-agentfield-adapter/fixtures/agentfield_responses/grn_experiment_complete:
```

Dryrun smoke

```
sudo config --target aiengineer bootstrap step run_paperclip_agentfield_adapter_dryrun_smoke

find /workspace/runs/paperclip-agentfield-adapter/dryrun -name agentfield_request.json | tail -n 1
find /workspace/runs/paperclip-agentfield-adapter/dryrun -name paperclip_status.json | tail -n 1
find /workspace/runs/paperclip-agentfield-adapter/dryrun -name paperclip_card.json | tail -n 1
find /workspace/runs/paperclip-agentfield-adapter/dryrun -name review_actions.json | tail -n 1
```

Mapping correctness test

For the POC fixture, agentfield_request.json must contain:

```
input.name = GRN Discovery in Human Cortex
input.dataset_ref = GSE123456_cortex_scrna
input.organism = human
input.method_flags = ["pca", "correlation", "perturbation"]
```

Status mapping test

Mock completed response should map to:

```
phase = completed_review_required
selected_agents present
stage_summaries present
final_summary present
review_actions present
```

Live smoke preflight

Before live smoke:

```
curl http://localhost:8080/health
```

or adapter equivalent health check.

Then:

```
sudo config --target aiengineer bootstrap step run_paperclip_agentfield_adapter_live_smoke
```

Expected:

```
agentfield_response.json exists  
paperclip_card.json exists
```

Scientific / platform guardrails

Bundle 12 must not turn Paperclip into the experiment engine.

Wrong:

```
Paperclip directly runs NCA training.  
Paperclip directly reads/writes nca-art-grn internals.  
Paperclip decides scientific truth.
```

Correct:

```
Paperclip requests/reviews.  
Adapter translates.  
Agentfield orchestrates.  
nca-art-grn executes science.  
Research Scientist approves meaning.
```

Also, Paperclip should not show “success” just because an LLM produced a confident summary.

The card should distinguish:

workflow completed
human review required
scientific evidence incomplete
artifact missing
platform failure
science failure

Summary

Bundle 12 is the **Paperclip-Agentfield integration layer**.

It turns this:

Paperclip job/request

into this:

Agentfield run_experiment input

and turns this:

Agentfield status/results

into this:

Paperclip-visible card, artifact links, and review actions

Platform-wise:

config prepares adapter
adapter maps Paperclip <-> Agentfield



Agentfield runs experiment-aware controller
nca-art-grn later performs science
Paperclip later exposes the human workflow

Research-wise:

Bundle 12 makes experiment-aware GRN workflows reviewable by a human,
without letting the dashboard replace the science engine or the controller.